

Possibilities for Designing Enhancing Spatial Knowledge Acquirements Navigator

A User Study on the Role of Different Contributors in Impairing Human Spatial Memory During Navigation

Jie Lu

Beijing Engineering Research Center of Mixed Reality and Advanced Display, School of Optics and Photonics, Beijing Institute of Technology, Beijing, China
lll5.jie@gmail.com

Yu Han

Beijing Engineering Research Center of Mixed Reality and Advanced Display, School of Optics and Photonics, Beijing Institute of Technology, Beijing, China
han.yu@outlook.com

Yunfei Xin

Beijing Engineering Research Center of Mixed Reality and Advanced Display, School of Optics and Photonics, Beijing Institute of Technology, Beijing, China
nolan58@163.com

Kang Yue

Beijing Engineering Research Center of Mixed Reality and Advanced Display, School of Optics and Photonics, Beijing Institute of Technology, Beijing, China
einhep@bit.edu.cn

Yue Liu

Beijing Engineering Research Center of Mixed Reality and Advanced Display, School of Optics and Photonics, Beijing Institute of Technology, AICFVE of Beijing Film Academy, Beijing, China
liuyue@bit.edu.cn

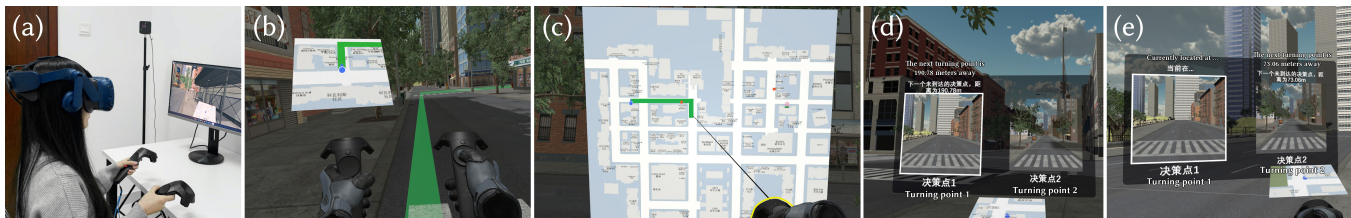


Figure 1: An overview of this study. (a) participants with the HTC Vive Pro; (b) Mode 1: Effective and Normal, system planned routes; (c) Mode 2: Return Route Decision, participants decided the route; (d) Mode 3: Re-spread participants' Attention to Surroundings, participant summoned the scroll-window; (e) Mode 3: Re-spread user's Attention to Surroundings, participant summoned the scroll-window reached a turning point.

ABSTRACT

Satellite mobile navigation systems have been criticized for impairing human spatial memory as an assistant in daily life. Nevertheless, few studies have directly investigated the specific role of different main contributors in such impairment in the same context. This study selected two of the main contributors leading to the impairment of human spatial memory and then made every possible effort to eliminate them from the navigation system: return user's route

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decision-making control, and subliminally return user's attention that has ever been distracted by the system to the environment. Two navigation modes were designed and compared in detail with the different spatial knowledge acquired by the efficient everyday navigation system. A within-subjects study in Virtual Reality (VR) urban environment was conducted and data on spatial knowledge was collected and analyzed. We concluded that the redirection of attention significantly benefits the acquisition of landmark as well as route knowledge, and the increase of route decisions further benefits survey knowledge as well as self-orientation. We also reflected on how these results might implicate future navigation systems design, and the results of analysis showed that it was possible to design a navigation system that does not impair or even enhance user's spatial memory while still retaining efficient navigation. Based on our findings, different future application-specific studies can better design navigation systems that focus on different spatial memories.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; User studies.

KEYWORDS

Spatial Knowledge, Navigation, Virtual Reality, Mobile Device

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1 INTRODUCTION

Powered by advanced transportation technology and the tendency of globalization, people currently are often faced with the need to travel to unfamiliar areas. Satellite mobile navigation systems have become an inseparable part of modern daily life in this context. Despite the popularity of efficient navigation systems, concerns have also been raised in the pieces of literature about the adverse effects on spatial knowledge acquisition caused by their extensive use [1]. However, such kinds of GPS-based mobile navigation systems still suffer from unreliable GPS connection in densely populated urban areas and an existing broad range of works reported that navigation aids could reduce the ability to learn new environments and impair spatial memory formation [2, 3].

Using a navigation system that does not impair the learning of a new environment or even enhances this process can implicitly develop good navigation habits and spatial orientation skills in the process of using it. On the other hand, when the current navigation system is not available, users can navigate themselves based on their own intact mental configuration of the spatial environment they have formed and can reduce the anxiety as well as the confusion of being in an unfamiliar environment. Actually, the mental configuration of the map affects every aspect of our lives, not only when meeting an emergency but also when being a novice in a 3D game and traveling.

Cushman et al. [4] have shown a correlation in navigation performances between real and virtual environments [5]. Considering that prior knowledge might be a confounding factor, and with the pandemic of coronavirus, it is difficult to find both safe and unfamiliar environments that none of the participants have ever known. Virtual Reality (VR) environment can satisfy this need because it is highly controllable in terms of variables, as well as cheaper and safer to monitor scientifically.

In this paper, three different navigation aided tools (modes) were compared on three non-overlapping but comparable routes in the same VR scenario, Figure 1 provides an overview. Our aim is to find out the specific effects of different factors on user's spatial memory when these factors are directly and simultaneously compared so that subsequent research can take these results and design a more targeted system to enhance user's navigation experience according to their needs. We first compared the spatial knowledge acquired in these three presented navigation modes, then give some reasonable

suggestions for future navigation systems after analyzing collected data.

2 RELATED WORK

Spatial knowledge of places develops in a sequence of three steps according to a widely accepted spatial learning model. First is landmark knowledge, recognition and recall of landmarks in the environment. The second is route knowledge, the ability to navigate along a route without getting lost. The final element is survey knowledge, knowledge of the orientation and distance between locations in the environment, survey knowledge doesn't repel detouring and shortcutting. Not everyone has detailed survey knowledge of their surroundings; however, exposure to maps of places clearly would advance the development of such knowledge [6, 7].

In order to reduce spatial memory impairments during navigation, it is necessary to understand the specific reasons behind these impairments and how they work underneath. Prior studies have concluded specific explanations for this spatial memory impairment during modern navigation. Some researchers mentioned that decreased environment engagement is the main contributor [8, 9]. In addition, other effects caused by navigation systems, such as lack of spatial decision making [10] and divided attention [11], were also considered as contributing factors. These three factors have been regarded as the main contributors that impaired spatial memory [12]. Nevertheless, few studies have directly compared different factors at the same time and analyzed spatial knowledge.

Freitag et al. [13] emphasized the importance of spatial knowledge, proposed automatic assistance for recommending some important and unvisited places to users in a free exploration context, preventing the incomplete environment knowledge owing to missing essential parts. Similarly, Colby et al. [14] illustrated the importance of map knowledge in 3D games for novice experience. They confirmed that navigation assistance in games might not impair user's mental map formation by investigating participants' relevant spatial knowledge. However, their application scenarios were mainly focused on 3D games and did not investigate the effects of specific factors in everyday life. In contrast, our study is not only applicable to everyday navigation but can also be extended to gaming scenarios as appropriate. Bertel et al. [15] analyzed spatial knowledge acquired from the tactile and visual conditions in detail. For designing a better navigation system, Gardony et al. [16] investigated how users' strategies interacted with the AR model and individual differences could yield the best spatial memory and navigation performance in a study of VR accident back-to-origin navigation tasks. Gardony et al. [12] cited dividing attention as a more contributing factor rather than selective interference of verbal working memory during a study of navigation in VR environment with verbal, tonal aid or without aid, however, they did not directly compare these two factors.

Dominic and Robb [17] compared the effect of different types of virtual annotations and different levels of navigation information on performance during a navigation task in VR and they found that world-fixed annotations with a higher level of information performed best in both time-taking and travel distance tasks. As concluded in [17], a world-fixed annotation and a small map with

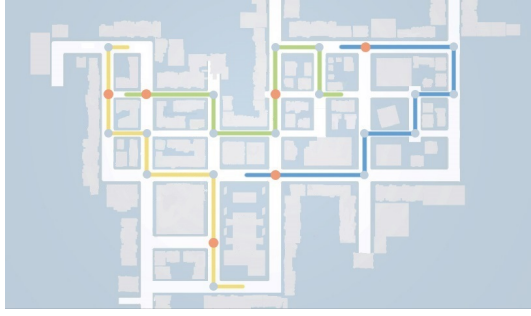


Figure 2: Three routes for navigation task. The grey point indicates a turning point, the orange point indicates a point without notification in Mode 3.

a planned path could simulate the efficient navigation systems in such work.

3 METHOD

3.1 Study Design

A within-subjects study was used to compare three different navigation modes; one is regular, efficient, yet the other two took account of previously demonstrated factors that could impair acquiring spatial knowledge, that is, tried to increase the spatial decision making, redirected attention back to the surrounding environment. Correspondingly, three non-overlapping routes have been elaborately selected and designed in the same VR scenario (See Figure 2); each is 1.4km long with eight decision points (e.g., intersection) and six turning points, which will be discussed in navigation modes. To account for potential order effects, pairings and orders of the condition were balanced across participants. Each participant traveled along all three routes with different navigation modes in three times, respectively. Similar to work in [15], we supposed that daily life navigation involved dual tasks, i.e., before arriving at the destination, users were allowed to free explore the buildings along the route.

Since individual spatial skills might be a confounding factor [16], this study involved two spatial ability tests: a questionnaire about the sense of direction, Santa Barbara Sense of Direction Scale [18], and an objective measure, Perspective Taking/Spatial Orientation Test (PTSOT) to measure orientation skills [19, 20].

A total of 18 participants (12 male and 6 female) recruited from a university took part in the study. Ages ranged from 20 to 28 years ($M = 22.89$, $SD = 1.81$). Of the 18 participants, 15 (83.3%) had no previous experience with VR, 2 (11.1%) had some previous experience with VR, and 1 (5.6%) declared themselves to be familiar with VR. No participants reported elevated susceptibility for motion sickness when queried using the Motion Sickness Susceptibility Questionnaire Short-form (MSSQ-Short) [21]. All participants were provided written informed consent before the experiments.

3.2 Apparatus

A virtual urban scene was developed by Unity, which occupied 495,000 m² and included 204 buildings such as banks, malls, restaurants and so on. Some atmosphere sound effects were added to

better simulate the environment. A powerful desktop PC (Intel Core i7 @ 3.8 GHz, NVIDIA RTX 2070s, 32 GB of RAM) was used to run the virtual scene. And the HTC Vive Pro headset with two controllers was chosen for interacting with the virtual environment. The left-hand controller was used to control straight movement in the environment. Moreover, a smooth turn, which enables participants to track surrounding objects while turning, to better simulate human walking, was supported by the click on a right-hand controller. The experiment was performed in a distraction-free experimental room (size 3 m × 3 m), in a comfortably seated situation to reduce simulator sickness (See Figure 1 a).

3.3 Three Different Navigation Mode

3.3.1 Mode 1: Effective and Normal (EN-Mode 1). We applied a planned route as a world-fixed annotation in the virtual environment (See Figure 1 b). In order to simulate the commonly used navigation-aided mode in the real world, we also provided a mini-map with a planned-path on which users can easily zoom.

3.3.2 Mode 2: Return Route Decision (RRD-Mode 2). The path planning control was entirely returned to the users, users needed to plan the route on their own to eliminate the “lack of spatial decision making” factor. The indicator bounded with the right-hand controller performs path planning on the big map, which always pointed to the north (See Figure 1 c). Some sections of the map were marked as under repair to avert different path effects. Under such circumstances, all routes with the same destination provided by the different aided modes were the same.

3.3.3 Mode 3: Re-spread user’s Attention to Surroundings (RAS-Mode 3). The attention of user was subliminally transferred back to the surroundings by this aided tool. The trigger key of the left-hand controller could summon all “Turning points photos” of the route in the same sliding window, which was designed for indicating the number of turning points while enhancing their unconscious impression of essential spatial knowledge of the entire route. The so-called “Turning points photo” is an image captured by the system at a turning point looking at the next turning point so as to denote the turn information; if there is a crossing that needs to go straight directly, then this point is not a turning point. The users needed to compare the critical information abstracted from this photo with their surroundings so their attention could be brought back to this environment. Also, the text above the photo would indicate their positions relative to these turning points (See Figure 1 d, e). A soft vibration of the right-hand controller will occur when a turning point is closed.

Participants were provided with a mini-map adapted to user’s orientation indicated by a user icon among three conditions, which has a maximum and a minimum size, and the way users moved through the environment would not differ.

3.4 Procedure

After filling in all the forementioned questionnaires, the participants were briefed on how to interact with the navigation tool they need to use in this condition and then wore the headsets with the help of an experimenter.

Table 1: Key results from statistical analysis

Dependent Variable (Scores)	Statistical Test	Test Statistic	<i>p</i>	η^2
Photo Selection	Kruskal-Wallis	H = 12.446	0.002*	0.205
Photo Sequence	ANOVA	F = 4.489	0.016*	0.150
Map-Drawing	Kruskal-Wallis	H = 15.763	0.000*	0.270
Pointing Error	Kruskal-Wallis	H = 15.428	0.000*	0.263
Decision Points Selection	ANOVA	F = 3.060	0.056	-
Decision Points Turn	Kruskal-Wallis	H = 5.153	0.076	-

* of $p < 0.05$

At first, the subjects familiarized themselves with the equipment and the navigation mode within 5 minutes in a simple VR environment. When they considered themselves comfortable and free to go, the experiment started. As they first arrived at the start point, the experimenter told them to memorize this position. With our dual-task assumption, each route had two defined additional has-to-passes by landmarks marked as a red dot on the maps. When passing, the experimenter would inform participants to memorize its location. After they achieved their navigation destination, participants were then asked to face north and recall their starter position and the other two passing by relative to north. Once they took off the headsets, they needed to draw these three directions as they did in PTSOT before. A test for their spatial knowledge was followed; ten photos taken in the environment were given, six of which correctly showed the places they passed by. There existed five photos of crossing points in this test which not always followed the direction of walking; four had been seen along the route. During the test, participants needed to pick right passed-by photos; if a photo of crossing was determined to belong to the correct answer, they also needed to indicate where they come from and where to go next, then they were asked to order the picked photos. They were allowed to select any number of photos the whole time.

The above process was repeated for every route, and each condition was carried out at least one day apart so as to avoid overloading and ceiling effect.

4 RESULTS

All participants completed the main navigation task in time. We have studied the impact comparison of proven negative contributors during navigation using VR technology. The first purpose of this study was to investigate all spatial knowledge acquired when doing our utmost to eliminate these factors. The second purpose was to identify which factor is more significant for concluding guidelines for future work.

The Shapiro-Wilk test of normality distribution was used to examine the distribution of collected data; for those that followed a normal distribution and satisfied equality of variances, a one-way ANOVA was applied, which was followed by a Student's *t*-test for post-hoc analysis; for the others, a Kruskal-Wallis test was used (See Table 1), which was followed by a Wilcoxon signed-rank test for post-hoc comparison. The effect size η^2 was used to estimate the practical significant difference between groups [22].

4.1 Analysis of Landmark Knowledge

To assess landmark knowledge gains of three navigation mode, the scores of photo selection were computed; each hit was scored as +1, and each miss as -1. A Kruskal-Wallis test detected a significant difference among the three conditions ($p = 0.002$, $\eta^2 = 0.205$). Post-hoc pairwise comparisons revealed that participants acquired more landmark knowledge in the RAS-Mode 3 condition than all other conditions (EN-Mode 1 with $p = 0.069$), i.e., when the system correctly guiding participants' attention to the environment, participants markedly gained more landmark knowledge. However, the RRD-Mode 2 condition performed better than the EN-Mode 1 condition (EN-Mode 1: $M = 1.780$, $SD = 1.987$; RRD-Mode 2: $M = 2.220$, $SD = 1.263$; RAS-Mode 3: $M = 3.56$, $SD = 0.856$).

4.2 Analysis of Route knowledge

We used sequence score to denote route knowledge, whereas two correct (incorrect) sequential photos were scored as +1 (-1). The performed ANOVA suggested a statistical significant difference among the three modes ($p = 0.016$, $\eta^2 = 0.150$). Further Bonferroni-adjusted planned pairwise comparisons revealed that RRD gained more route knowledge than NE ($p = 0.019$). No significant differences emerged for the other pairwise comparisons (all p 's > 0.1). A trend towards significance was observed ($p = 0.089$) between RRD and RAS, and participants acquired most route knowledge after redirecting their attention back to the environment over all conditions (EN-Mode 1: $M = -3.720$, $SD = 5.696$; RRD-Mode 2: $M = -2.670$, $SD = 5.087$; RAS-Mode 3: $M = 1.220$, $SD = 4.821$).

4.3 Analysis of Survey knowledge

We selected a map-drawing score to characterize participants' survey knowledge. This was scored by the similarity between the drawing route and the correct one. A Kruskal-Wallis test yielded a significant difference among three conditions ($p = 0.000$, $\eta^2 = 0.270$). Post-hoc pairwise comparisons revealed that participants performed significantly better in the RRD-Mode 2 condition than in the EN-Mode 2 and RAS-Mode 3 conditions (EN-Mode 1: $M = 0.550$, $SD = 0.391$; RRD-Mode 2: $M = 0.933$, $SD = 0.154$; RAS-Mode 3: $M = 0.672$, $SD = 0.290$). This is totally reasonable, although we told participants that they were free to use the map anytime in the RRD-Mode2, some chose to memorize them first and then started their navigation, for others, every time they opened the map, they have formed a better survey knowledge. Moreover, the performance in RAS-mode 3 condition is slightly better than EN-Mode 1 condition.

4.4 Analysis of Direction Estimates

The error in participants' pointing toward their three memorized location was computed for each trail and was converted to a sum of absolute pointing error for analysis. A Kruskal-Wallis test was conducted and the result showed that the direction estimates was significantly different ($p = 0.000$, $\eta^2 = 0.263$) across the three conditions. Post-hoc pairwise comparisons revealed that there was a significant difference between RRD-Mode 2 and RAS-Mode 3 (EN-Mode 1: $M = 147.136^\circ$, $SD = 137.181^\circ$; RRD-Mode 2: $M = 66.960^\circ$, $SD = 104.515^\circ$; RAS-Mode 3: $M = 197.353^\circ$, $SD = 128.145^\circ$). An analysis of pairwise correlations between the two spatial abilities scores - SBSOD and PTSOT - was performed, and results of the pointing tasks revealed no significant relationships.

4.5 Analysis of Navigation Decisions

Decision-making is an essential attribute of navigation. The second mode used a map to maximize the level of control and the amount of decision-making. Our third mode used a sequence of photos captured at our "turning points" that suggested participants unconsciously follow the features of their surroundings. So, it's interesting for us to investigate the participants' performance on decision-making. We computed the score of decision points selection as the rule of photo selection, and for turn scores, correctly (incorrectly) indicated turns at the right position were scored as +1 (-1).

A one-way ANOVA was conducted to analyze the decision-points selection scores, and a trend towards significance was observed ($p = 0.056$). Participants performed the best when using photos; surprisingly, when participants made their own decisions, their performance was slightly worse than the usual navigation (EN-Mode 1: $M = 0.22$, $SD = 4.278$; RRD-Mode 2: $M = -0.11$, $SD = 3.787$; RAS-Mode 3: $M = 2.56$, $SD = 2.148$). A possible explanation is that the amount of attention paid to the map in Mode 2 was similar to the planned path in Mode 1.

For turn scores, a trend towards significance was observed ($p = 0.076$) by a Kruskal-Wallis test. However, according to the way we designed the third mode, we supposed that we would get a significant difference among the three conditions (EN-Mode 1: $M = -0.89$, $SD = 4.129$; RRD-Mode 2: $M = -1$, $SD = 4.298$; RAS-Mode 3: $M = 1.33$, $SD = 3.068$). A possible explanation is the disorientation in the VR environment [23]. Another explanation is the way we designed the third mode, to keep the three navigation modes separated, only a small local map was applied in these three modes except for mode two so that users were not able to know their geographic orientation in a global map very clearly. One participant reported that "When making decisions about intersections, my memory of instructions such as 'left, right' was lost by judging the direction of the turn through the image."

5 IMPLICATIONS FOR DESIGN AND FUTURE WORK

A few participants noted that "The way I navigate in daily life is basically to just look at the navigation paths on my phone and places under my feet." Some reported that after this experiment, they realized a common navigation system did impair their spatial knowledge

acquirements. A specific participant said, "Mode 3 may be appropriate for traveling." Moreover, several participants mentioned they felt a little mentally overloaded with Mode 2. In line with previous studies, emphasizing user's decision-making by making them proactively plan the route and enhancing user's involvement in the environment does have a positive effect on the formation of spatial knowledge during navigation. However, the focus of the two was not consistent, as improving user's decision-making can increase user's spatial knowledge acquisition in the survey perspective, which could significantly improve user's orientation in unfamiliar environments. In contrast, reducing distraction or redirecting attention has a positive effect on both landmark and route perspective. Whether the reason for no significant difference observed in the direction-estimating task is the way we directed user's attention or the specificity of the VR environment [23] requires further research and consideration. Thus, a future study could design more tricks about how to increase user's making decision on the route to attach more significance to survey perspective, just as 2D or 3D games played in a bird view. Another possibility for future research is that designers could start with user's attention to develop a navigator focusing on landmark and route perspective. For adaptive navigation systems, users can fill in some questionnaires to indicate their habits of perspectives, and then if they have already had a good development of route perspective, the system could make them pay more attention to the surroundings to improve their navigation skills.

The way we directed user's attention in Mode 3 can be easily merged in the existing navigation methods using panoramic photos. In this paper, we did not provide a global overview of the entire environment in Mode 1 and 2 in order to strip away the different contributors as much as possible. Actually, we could enhance user's sense of direction by designing a small pop-up window in the interface showing user's position on the global map when reaching a turning point or some crucial points. However, the primary purpose of this study is to accurately identify the differences between the two contributors in spatial knowledge acquisition, and provide implications for future work. In the future design of navigation systems that improve user's spatial knowledge acquisition, other effective and novel ways of directing user's attention can be developed (e.g., link key features or views with unique things by slightly increasing user's own route planning; or adaptively display key information along the route according to its importance). It is possible to design a navigation system that forms a more complete knowledge of the environment under the right mental load for the users without compromising the efficiency of reaching the destination, even improves the ability to learn a new environment.

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